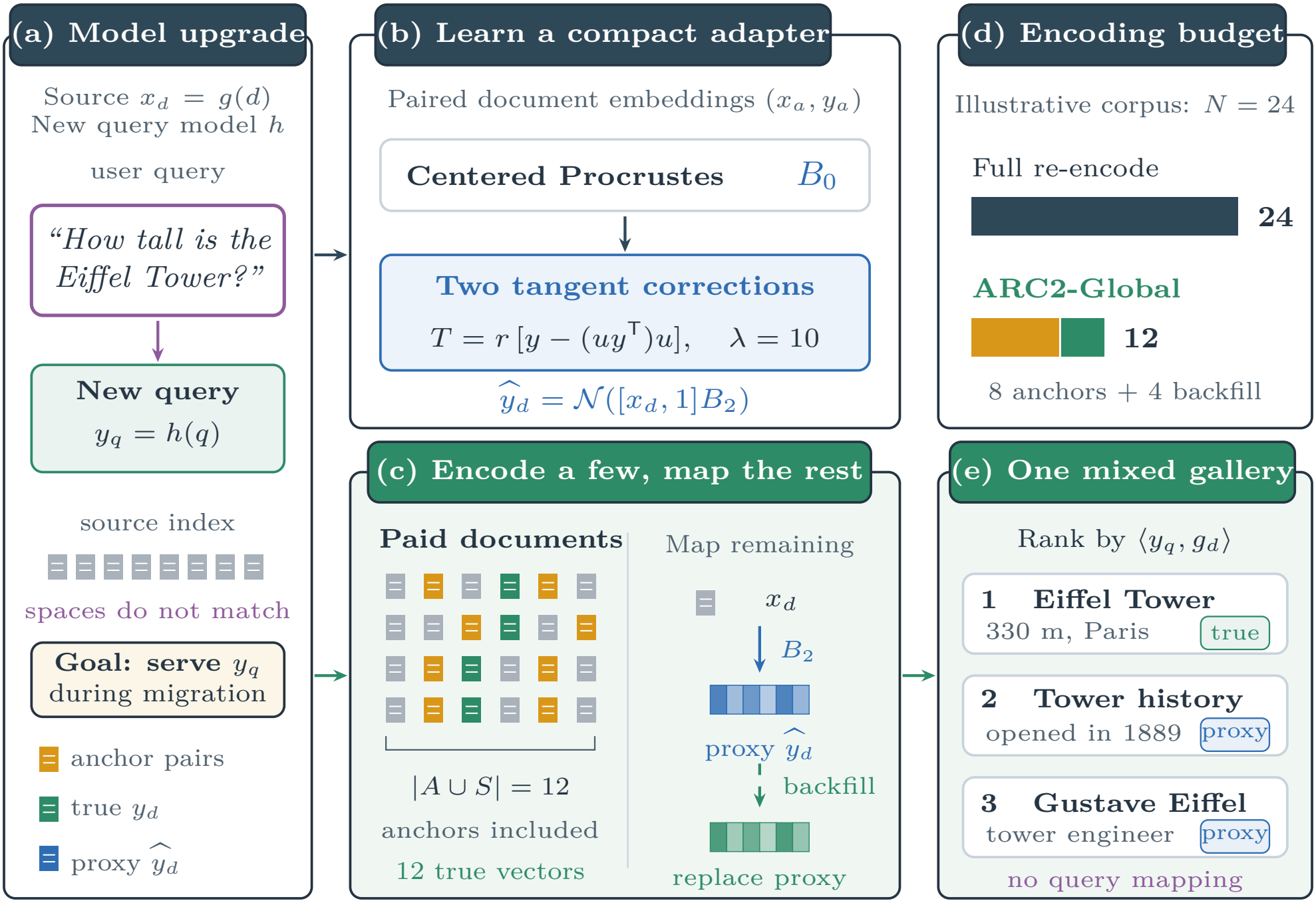

(a) Model upgrade
Source $x_d = g(d)$
New query model h
user query
"How tall is the Eiffel Tower?"
New query
$y_q = h(q)$
source index
spaces do not match
Goal: serve y_q during migration
anchor pairs
true y_d
proxy $\widehat{y}_d$
(b) Learn a compact adapter
Paired document embeddings (x_a, y_a)
Centered Procrustes B_0
Two tangent corrections
$T = r\,[y - (uy^{\mathsf{T}})u], \quad \lambda = 10$
$\widehat{y}_d = \mathcal{N}([x_d, 1]B_2)$
(c) Encode a few, map the rest
Paid documents
$|A \cup S| = 12$
anchors included
12 true vectors
Map remaining
x_d
B_2
proxy $\widehat{y}_d$
backfill
replace proxy
(d) Encoding budget
Illustrative corpus: $N = 24$
Full re-encode
24
ARC2-Global
12
8 anchors + 4 backfill
(e) One mixed gallery
Rank by $\langle y_q, g_d \rangle$
1 Eiffel Tower
330 m, Paris
true
2 Tower history
opened in 1889
proxy
3 Gustave Eiffel
tower engineer
proxy
no query mapping